MILITARY SPECIFICATION

FILTER ELEMENT AND RING ASSEMBLY

1. SCOPE

1.1 This specification covers a filter element and ring assembly for assembly into canisters.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Locked-Corner.
PPP-B-636 - Box, Fiberboard,
PPP-F-320 - Fiberboard; Sheet, Stock and Cut Shapes.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive paper, (For Carton Sealing).

FSC 4240

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

D5-3-892 - Filter Element and Ring Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein, Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-3-892 or on subsidiary drawings shall conform to the specifications listed thereon or the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the filter element and ring assembly shall conform to the specifications and the drawings listed on Drawing D5-3-892 and subsidiary drawings.

3.2 Manufacture and assembly. The filter element and ring assembly shall be manufactured and assembled as specified on Drawing D5-3-892-1 or D5-3-892-2 as specified in the contract or order (see 6, 2).

3.3 Air resistance. The air resistance of the filter element and ring assembly shall not exceed 10 millimeters of water at an airflow rate of 32 liters per minute when tested as specified in 4.6.1.

3.4 Dioctyl phthalate (DOP) penetration. The filter element and ring assembly shall not al low more than 0.015 percent DOP penetration at a flow rate of 32 liters per minute *when* tested as specified in 4.6.2.

3.5 Adhesion. The filter material shall be thoroughly and complete} y bonded with the paper tab at all points of contact and with itself at the center pleat. The filter material shall tear within itself rather than pulling loose, and filter material fibers shall remain securely attached to the adhesive when tested as specified in 4.6.3.

3.6 Preproduction. When specified in the contract or order, a preproduction sample shall be produced for examination and test. The preproduction sample shall be manufactured using the same materials, methods and equipment as will be used during regular production (see 6.2).

3.7 Workmanship. The filter element and ring assembly shall be free from foreign matter, such as dirt, oily or viscous material, and free of damage, such as tears or punctures in the element or burrs, dents, or deformations in the ring assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible e for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the 'Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence, The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3, 1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Substation inspection. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL spcified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the filter elements and ring assemblies of one type produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

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4.4 Sampling.

4.4.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. Samples for air resistance and DOP penetration shall be selected from items unit packaged.

4.4.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-3.

4.5 Inspection provisions.

4.5.1 Examination and nondestructive tests. Sample filter element and ring assemblies shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.5.2 For destructive tests. Sample filter element and ring assemblies shall be tested in accordance with 4.6.3. The acceptance number is zero.

4.5.3 Classification of defects.

4.5.3.1 Filter element and ring assembly (Dwg D5-3-892).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component incorrectly assembled	
102	Component missing	
103	Overall diameter incorrect	
104	Overall height incorrect	
105	Workmanship (3.7)	
Major A:	AQL 0.25 percent defective	
150	Air resistance	4.6.1
151	DOP penetration	4.6.2
Minor:	AQL 2.5 percent defective	
201	Height outside fold incorrect	

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4.6 Tests.

4.6.1 Air resistance. The airflow resistance shall be determined using Indicator, Resistance, Q112.

4.6.2 DOP penetration, The DOP penetration shall be determined using Penetrometer, Filter Testing, DOP, Q127.

4.6.3 Adhesion. Tear the assembly apart by applying tension at al l cemented-d examine for compliance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging levels shall be as specified (see 6.2).

5.1.1 Level B.

5.1.1.1 Unit packaging. The filter element and ring assembly shall be individually wrapped in tissue or glassine paper.

5.1.1.2 Intermediate packaging. Five hundred filter element and ring assemblies shall be packaged in a type 1, class 2, grade 5, style RSC fiberboard box conforming to PPP-B-636. The assemblies shall be arranged in 20 layers, each containing 5 rows of 5 filter element and ring assemblies. The assemblies shall be separated from one another by fiberboard interlocking partitions and fiberboard pads between layers using the same type of fiberboard as the box or using fiberboard conforming to type CF, domestic class, variety SW of PPP-F-320. Closure and reinforcement shall be as specified for interior boxes of the box specification. Box corners, seams, and manufacturers joint shall be taped as specified in the box specification, except that pressure sensitive tape conforming to PPP-T-76 shall be used. The contents of the package shall be snug enough to prevent free movement of the filter element and ring assemblies without applying pressure which might deform them.

5.1.2 Level C. The filter element and ring assemblies shall be packaged as specified in 5. 1.1 except the fiberboard box shall be type 1, class 1, style RSC box conforming to PPP-B-636.

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5.2 packing. Levels shall be as specified (see 6.2).

5.2.1 Level B. Filter element and ring assemblies packaged as specified in 5.1 sha-eked in a style 4 or 4-1/2 nailed wood box conforming to class 1 of PPP-B-621, Gross weight of the. box shall not exceed the weight l imitations of the box specification,

5.2.2 Level C. Filter element and ring assemblies packaged *as* specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at first destination for immediate use. Containers used shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Unless Otherwise specified in th$_e$ contract or order, unit and intermediate packages and shipping containers shot l be marked in accordance with MIL-STD-1290

6. NOTES

6.1 Intended use. The filter element and ring assembly covered in this specification is a component of the M14 particulate canister.

6.2 ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of filter material to be used.
(c) Level of packaging and packing required.
(d) Whether the D5-3-892-1 or D5-3-892-2 assembly is required.
(e) Whether a preproduction sample is required.

(1) Quantity required,

(2) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(3) Name and address of test focility and shipping instructions when testing is performed by the Government.

(4) Time required for the Government to notify the supplier whether or not to proceed with production. .

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6.3 Adhesive. An adhesive suitable for cementing the filter has been prepared according to the following formulation:

Ingredients	Parts by weight
Latex, 60-percent, centrifuged (Firestone Rubber & Latex Co.)	810
Ten percent solution NaOH	20
Ten percent solution Alkanol (Dupont)	25
Forty percent solution formaldehyde	20
*Vulcanizing Dispersion C95 (R. T. Vanderbilt Co., Inc.)	50

*Added just prior to use of cement.

6.4 Number of pleats. When relatively thick filter material is used in production of the filter element, it is generally advantageous to use a lower number of pleats in order to meet the air resistance requirement When relatively thin material is used, it is generally advantageous to use a higher number of pleats in order to meet the DOP penetration requirement.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 4240-A196

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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